Transforming Graphs using Logs

Exam code: 9MA0

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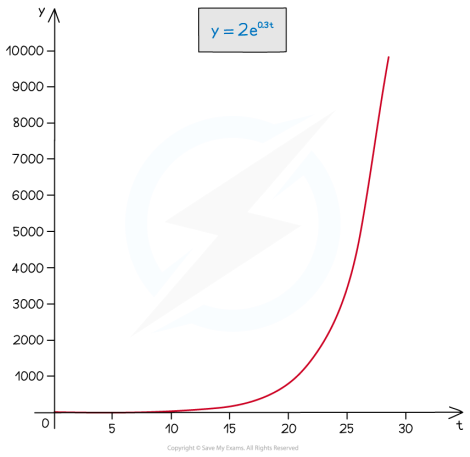
Transforming graphs using logs

What are logarithmic axes?

- For graphs that increase or decrease exponentially
  - it can be very difficult to read specific values from the scales on the axes



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$y = 2e^{0.3t}$

$\ln y = \ln(2e^{0.3t})$

$\ln y = \ln 2 + \ln e^{0.3t}$

$\ln y = \ln 2 + 0.3t$

$\ln y = 0.3t + \ln 2$

TAKE LOGS OF BOTH SIDES

$\ln xy = \ln x + \ln y$

THIS IS OF THE FORM "y=mx+c"
"y" = ln y
"m" = 0.3
"c" = ln 2

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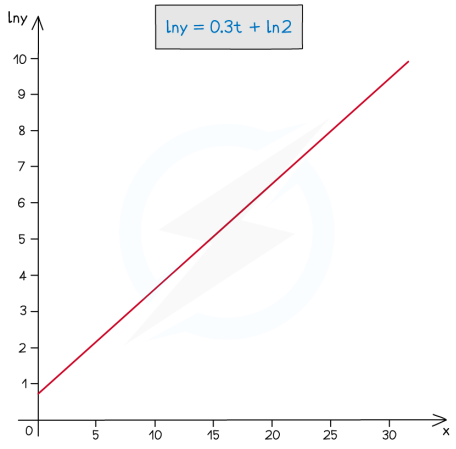
- Plotting **ln y** against **t** produces a straight line
  - The y-axis is a **logarithmic axis**



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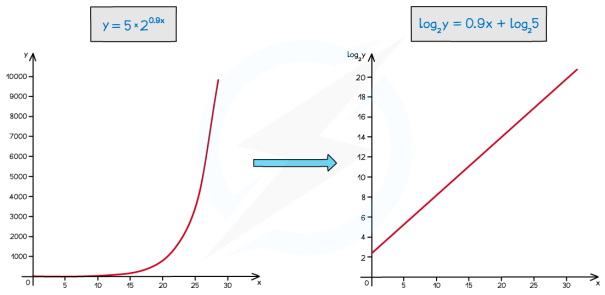
- Note the second graph has **ln y** on the **y**-axis
  - Log graphs have at least one logarithmic axis
- Reading a value for **ln y** at **t = 20** is easier than reading the value for **y**
- Logarithmic axes are used where a wide range of numbers can occur
  - it makes numbers smaller and easier to deal with
  - a curve can be turned into a straight line



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- Exponential models are of the form
  - **$y=Ab^{kx}$ (growth)**
  - **$y=Ab^{-kx}$ (decay)**

- To use a model to make predictions the values of **A**, **b** and **k** are needed
- **A**, **b** and **k** will usually be estimated from observed data values
  - **A** may be known exactly, as it is a starting/initial value

- Estimating from a straight line graph is easier than from a curve

TYPE 1

$y = Ab^{kx}$

BASE b (WHERE b>0 IS A CONSTANT)

$\log_b y = \log_b Ab^{kx}$

$\log_b y = \log_b A + \log_b b^{kx}$

$\log_b y = \log_b A + kx\log_b b$

$\log_b y = kx + \log_b A$

"$y = mx + c$"

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TYPE 2

$y = Ae^{kx}$

BASE e

$\ln y = \ln Ae^{kx}$

$\ln y = \ln A + \ln e^{kx}$

$\ln y = \ln A + kx\cancel{\ln e}^{1}$

$\ln y = kx + \ln A$

"$y = mx + c$"

THIS IS A SPECIAL CASE OF TYPE 1

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TYPE 3

$y = Ax^b$

x IS THE BASE

$\log y = \log Ax^b$

$\log y = \log A + \log x^b$

$\log y = \log A + b\log x$

$\log y = b\log x + \log A$

"$y = mx + c$"

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e.g. GIVEN $y=Ae^{kx}$ AND THAT $\ln y = 3.2x + \ln 12$
FIND THE VALUES OF A AND k.

$3.2x = kx$

$k=3.2$

TYPE 2 $y=Ae^{kx}$

$\ln A = \ln 12$

$A = 12$

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